Exp. No: 1

Setting Up the Environment And Preprocessing The Data

Code:

```
# ========================================
# Titanic Dataset Preprocessing
# ========================================

# 1. Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder

# ========================================
# 2. Load Dataset
# ========================================
df = sns.load_dataset('titanic')

# Display first five rows
print(df.head())

# ========================================
# 3. Explore Dataset
# ========================================
print("\nDataset Information")
print(df.info())
```

```
print("\nStatistical Summary")
print(df.describe(include='all'))

print("\nMissing Values")
print(df.isnull().sum())

# ========================================
# 4. Handle Missing Values
# ========================================

# Fill missing Age values with median
df['age'] = df['age'].fillna(df['age'].median())

# Fill missing Embark Town values with mode
df['embark_town'] =
df['embark_town'].fillna(df['embark_town'].mode()[0])

# Drop Deck column (too many missing values)
df.drop(columns=['deck'], inplace=True)

# Drop any remaining rows with missing values
df.dropna(inplace=True)

# ========================================
# 5. Encode Categorical Variables
# ========================================

le = LabelEncoder()

# Encode binary categorical feature
df['sex'] = le.fit_transform(df['sex'])

# Encode embark town
df['embark_town'] = le.fit_transform(df['embark_town'])
```

```
# Drop unnecessary or duplicate columns
columns_to_drop = [
 'embarked',
 'class',
 'who',
 'alive',
 'adult_male',
 'alone'
]

df.drop(columns=columns_to_drop, inplace=True)

# ========================================
# 6. Define Features and Target
# ========================================

X = df.drop('survived', axis=1)
y = df['survived']

# ========================================
# 7. Split Dataset
# ========================================

X_train, X_test, y_train, y_test = train_test_split(
 X,
 y,
 test_size=0.20,
 random_state=42,
 stratify=y
)
```

```
# ========================================
# 8. Feature Scaling
# ========================================

scaler = StandardScaler()

numerical_cols = ['age', 'fare']

X_train[numerical_cols] =
scaler.fit_transform(X_train[numerical_cols])
X_test[numerical_cols] = scaler.transform(X_test[numerical_cols])

# ========================================
# 9. Final Output
# ========================================

print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape :", X_test.shape)

print("\nFirst Five Rows of Training Data")
print(X_train.head())
```

Output

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

Statistical Summary

	survived	pclass	sex	age	sibsp	parch
count	891.000000	891.000000	891	714.000000	891.000000	891.000000
unique	NaN	NaN	2	NaN	NaN	NaN
top	NaN	NaN	male	NaN	NaN	NaN
freq	NaN	NaN	577	NaN	NaN	NaN
mean	0.383838	2.308642	NaN	29.699118	0.523008	0.381594
std	0.486592	0.836071	NaN	14.526497	1.102743	0.806057
min	0.000000	1.000000	NaN	0.420000	0.000000	0.000000
25%	0.000000	2.000000	NaN	20.125000	0.000000	0.000000
50%	0.000000	3.000000	NaN	28.000000	0.000000	0.000000
75%	1.000000	3.000000	NaN	38.000000	1.000000	0.000000
max	1.000000	3.000000	NaN	80.000000	8.000000	6.000000

	fare	embarked	class	who	adult_male	deck	embark_town	alive	\

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
count	891.000000	889	891	891	891	891	203	889	891						
unique	NaN	3	3	3	2	7	3	2							
top	NaN	S	Third	man	True	C	Southampton	no							
freq	NaN	644	491	537	537	59	644	549							
mean	32.204208	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
std	49.693429	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
min	0.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
25%	7.910400	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
50%	14.454200	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
75%	31.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN							
max	512.329200	NaN	NaN	NaN	NaN	NaN	NaN	NaN							

	alone
count	891
unique	2
top	True
freq	537
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

Missing Values

survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

Training Data Shape: (711, 7)
Testing Data Shape : (178, 7)

First Five Rows of Training Data

	pclass	sex	age	sibsp	parch	fare	embark_town
621	1	1	0.967941	1	0	0.419754	2
481	2	1	-0.107871	0	0	-0.646083	2
527	1	1	-0.107871	0	0	3.851758	2
435	1	0	-1.183683	1	2	1.787603	2
797	3	0	0.122660	0	0	-0.469980	2